

Rik L.E.M. Ubaghs

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Bio

I am an engineer specializing in neurotechnology and optical systems, with expertise in biomedical imaging, medical devices, and data analytics. My work focuses on translating advanced experimental technologies into practical clinical applications, bridging the gap between cutting-edge research and real-world medical impact.

Education

ETH Zurich, Zurich, Switzerland <i>Ph.D. Neurotechnology</i>	July 2017 - May 2023
ETH Zurich, Zurich, Switzerland <i>CAS Entrepreneurial Leadership in Technology Ventures</i>	Mar 2025 - Mar 2026
University of Amsterdam, Amsterdam, the Netherlands <i>Research Master Brain and Cognitive Sciences, Cum Laude</i>	Sept 2014 - June 2016
Maastricht University, Maastricht, the Netherlands <i>Bachelor Psychology and Neuroscience, Cum Laude</i>	Sept 2011 - June 2014

Skills

Languages:	Dutch (Native), English (Fluent), German (Basic)
Programming:	Python, Rust, Matlab, R
Databases/Querying:	SQL, MySQL
Skills:	Data Science, Biomedical Imaging, Optical Systems, Mechanical Design (Autodesk, Solidworks)

Industry Positions

Arago Labs <i>Co-founder, CEO</i> - Founded an early-stage medical technology venture focused on improving surgical precision in oncology. - Designed advanced optical imaging platform combining nonlinear microscopy, and ML/DL-based tissue classification. - Defined market entry strategy, pricing model, reimbursement pathways, and go-to-market roadmap.	June 2023 - Current
Agora Technology, Maastricht, The Netherlands <i>Founder</i> - Built data solutions that aimed at streamlining/automating the workflow of medical facilities, especially focusing on surgical equipment. - We focused on internal logistics, and the creation, maintenance, and analysis of large-scale medical equipment databases using ML/DL.	Nov 2016 - Feb 2018

Research Positions

Inst. of Neuroinformatics, ETH Zurich, Zurich, Switzerland

July 2017 – June 2023

Ph.D. Candidate

- Development of a MRI compatible fluorescence microscope (full-cycle product design).
- Design of a paradigm to combine microscopy and MR imaging methods in awake rodents.
- Design of data pipelines for multi-modal data analysis.

Center for Neuroengineering, Duke University, Durham, USA

Dec 2015 – Nov 2016

Visiting Scientist

- Design of a hybrid brain decoding strategy in Non-human Primates.
- Development of ML/DL decoding pipeline for brain decoding.

Publications

Publications:

- **Ubaghs, R.L.E.M.**, et al. Simultaneous single-cell calcium imaging of neuronal population activity and brain-wide BOLD fMRI. *bioRxiv* (2023); accepted in *Nature Methods*.
- Rychen, J., ... **Ubaghs, R.L.E.M.**, ... Hahnloser, R. WhaleSonic: Recordings of cetacean vocal behaviour with hydrophone arrays in Northern Norway. *bioRxiv* (2023); submitted to *Scientific Data*.

Patents:

Sainz Villalba, L., **Ubaghs, R.L.E.M.** DEVICE AND METHOD FOR PROVIDING HISTOLOGICAL NAVIGATION INFORMATION DURING IN-VIVO SURGERY. European Patent Application, Filed 2026.